

Sleep Medicine: Insomnia and Sleep

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Abstract

Insomnia disorder is an economic burden and public health concern affecting up to one-third of the population of the United States. It is mostly seen in older age groups, and often considered a normal aging phenomenon. The diagnosis and treatment of insomnia rely mainly on a thorough sleep history to address the precipitating factors as well as maladaptive behaviors resulting in poor sleep. It is important for clinicians to recognize and manage the symptoms of insomnia to prevent the morbidity associated with it. This review aims to highlight the pathophysiology, associated comorbidities, clinical evaluation and effective management strategies for insomnia disorder.

Introduction

Insomnia is a public health concern and one of the most common complaints in medical practice. The disorder is characterized by difficulty with sleep quality, initiating or maintaining sleep, along with substantial distress and impairments of daytime functioning.¹ Studies have established insomnia to be a very common condition with symptoms present in about 33–50% of the adult population.² Its prevalence ranges from 10 to 15% among the general population, with higher rates seen among females, divorced or separated individuals, those with loss of loved ones, and older people.³ There is also an increased risk of depression, anxiety, substance abuse, suicide, motor vehicle accidents and possible immune dysfunction with chronic insomnia.⁴ Initially considered to be a symptom, insomnia is now defined as a disorder and classified separately in DSM-V (Diagnostic and Statistical Manual of Mental Disorders-5th edition) and ICSD-3 (International Classification of Sleep Disorders-3rd edition).

Pathophysiology

The genetic factors responsible for insomnia were identified from work on “insomnia-like *Drosophila* flies” (ins-I flies), which had traits similar to human insomnia. The genes associated with insomnia are Apolipoprotein (Apo) E4, PER3 (Period Circadian Regulator 3), Clock (Clock Circadian Regulator) and 5-HTTLPR (Serotonin Transporter Linked Polymorphic Region) genes. There is also a close association between insomnia and HLA-DQB1*0602.

The molecular factors responsible for the sleep-wake regulation include the wake-promoting chemicals like orexin, norepinephrine, and histamine, and sleep promoting chemicals like GABA (Gamma AminoButyric Acid), adenosine, melatonin, and prostaglandin D2. The orexin mediated increased neuronal firing in the wake-promoting areas (tuberomammillary nucleus, dorsal raphe and locus coeruleus) and inhibition of the sleep-promoting areas (ventrolateral preoptic nucleus and median preoptic nucleus) is one of the possible mechanisms contributing to insomnia (sleep switch model).⁵ Other possible mechanisms noted in the literature are briefly illustrated in [Figure 1](#).

Figure 1.

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Pathophysiology of insomnia (VLPO: ventrolateral preoptic nucleus; TMN: Tuberomammillary nucleus; DR: dorsal raphe; LC: Locus coeruleus; GABA: gamma-aminobutyric acid)

Classification

According to the International Classification of Sleep Disorders-3rd Edition (ICSD-III), insomnia disorder can be classified as:

1. Chronic Insomnia Disorder

Patient experiences sleep disturbances for the last three months affecting the night time sleep for at least three times a week.

2. Short-term Insomnia Disorder

Sleep disturbances experienced within three months.

3. Other Insomnia Disorder

Sleep disturbances that do not meet the criteria for chronic insomnia or short-term insomnia disorder are classified under this category.

The International Classification of Sleep Disorders, 2nd Edition describes the various subtypes of insomnia as:

1. Psychophysiological Insomnia

It is characterized by increased levels of cognitive and somatic arousal at bedtime. Such individuals describe excessive worrying about sleep along with difficulty with sleeping in their home environment. They may sleep easily in any other environment or when not planning to sleep.

2. Idiopathic Insomnia

It is characterized by sleep disturbances occurring early in childhood and persisting over a lifelong period. It may be associated with congenital or genetic variations in the sleep-wake cycle.

3. Paradoxical Insomnia

In this type of insomnia, the patients underestimate the total amount of sleep they obtained. They usually have a good night sleep which they perceive as the time of wakefulness. Paradoxical insomnia can be confirmed by polysomnography or actigraphy.

4. Inadequate Sleep Hygiene

Sleep hygiene highlights the effect of daily activity on the quality of sleep. Excessive daytime napping, evening consumption of alcohol or caffeine, watching television till late at night, working on electronic gadgets just before bedtime can negatively affect the sleep quality.

5. Behavioral Insomnia of Childhood

Insomnia in children may be affected by their dependency on certain stimulations, objects, environmental settings, disruption of which can result in significant delay in falling asleep (Sleep onset association type) or may show resistance to go to bed (Limit setting type) or both (mixed type).

Precipitating Factors

Although insomnia can affect any age group, women and elderly (>65 years) are the population more susceptible to the development of insomnia.

Psychosocial factors like the stress of work, shift work, loss of a loved one, divorce, domestic abuse can lead to significant sleep disturbances.

Developmental issues in children like delayed milestones, hyperactive behavior, separation anxiety can precipitate sleep disturbances in children.

Certain personality traits like excessive worrying, repressed personality, perfectionism, neuroticism can have a disturbing effect on sleep. Psychiatric comorbidities like depression, mood, and anxiety disorders, post-traumatic stress disorder can increase the risk of insomnia. Alcohol and substance abuse/dependence, excessive caffeine intake, excessive smoking can potentially affect the sleep-wake cycle.

Clinical Features

The sleep disturbances in insomnia can manifest as difficulty in falling asleep (Sleep Onset Insomnia), maintaining the continuity of sleep (waking up in the middle of the night and difficulty in returning to sleep) or waking up too early in

the morning well before the desired time, irrespective of the adequate circumstances to sleep every night (Early Morning Insomnia). Insomnia can significantly impact the daytime functioning resulting in waking up tired in the morning, decreased workplace productivity, proneness to errors and accidents, inability to concentrate, frequent daytime naps and poor quality of life.

In children, insomnia can be reported as frequent nighttime awakening, resisting to go to bed and sleep independently. Children may have a dependency on certain stimulations (rocking, storytelling), objects (bottle feeding, favorite toy) or room setting (parents in the room) to fall asleep, and lack of these stimulations can create anxiety and fear in them and result in sleep disturbances. Insomnia can affect their school performance, daily activity of playing, inability to concentrate and behavior problems.

Co-Morbidities Associated with Chronic Insomnia

Chronic insomnia disorder is a considerable risk factor for cardiovascular disease, hypertension, type 2 diabetes, gastroesophageal reflux (GERD) and asthma, the details of which are discussed under the following headings ([Figure 2](#)).

Figure 2.

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Comorbidities associated with chronic insomnia.

Insomnia and Cardiovascular Disease

Insomnia is a risk factor for cardiovascular morbidity and mortality.⁶ The underlying pathophysiology that explains this increased risk is mainly due to dysregulation of the hypothalamic-pituitary axis with increased release of adrenocorticotropin hormone, increased sympathetic nervous system activity, elevation of inflammatory cytokines and a rise in C-reactive protein level (CRP).^{7,8} Chronic insomnia is also noted to increase the risk of hypertension, reduce heart rate variability and increased atherogenesis.^{9,10} The HUNT Study noted a 27–45% increased risk of myocardial infarction in patients with chronic insomnia.¹¹ Even though the prospective data suggests a significant association of chronic insomnia with cardiovascular disease, further research is required to understand how the management of insomnia can impact the cardiometabolic health in these patients.

Insomnia and Type-2 Diabetes Mellitus

It is estimated that chronic insomnia increases the risk of type 2 diabetes mellitus (T2D) by 16%, in the adult population.¹² In a recent study by Lin et al., the risk of developing T2D was proportional to the duration of insomnia. They observed that in patients with chronic insomnia of <4, 4–8 and >8 years, the risk of T2D increased by 14%, 38%, and 51% respectively.¹³ The multiple mechanisms that might be involved in the pathogenesis include dysregulation of the hypothalamic-pituitary axis with an increase in the cortisol level, impairment in glucose metabolism, an imbalance in the leptin-ghrelin system that increases the appetite and risk of obesity resulting in insulin resistance and unstable blood sugars level.^{7,14}

Insomnia and Gastroesophageal Reflux Disease

A bidirectional association is noted between Gastroesophageal Reflux Disease (GERD), symptoms and sleep disturbances.¹⁵ In 2009, Mody et al., noted the effect of GERD on sleep quality. Out of 11,685 individuals with GERD, 88.9% experienced sleep disturbances, out of which 49.1% complained of difficulty in initiating sleep, and 58.3% had difficulty in maintaining sleep.¹⁶ In the same year, a cross-sectional cohort study by Jansson et al. on 65,333 patients with GERD observed, that there was three times increase in the risk of GERD in patients with insomnia.¹⁷ Further, the treatment of GERD with proton pump inhibitor has shown to improve the sleep disturbances in these patients significantly.¹⁸

Insomnia and Asthma

A potential risk of asthma and allergic rhinitis is noted in patients with chronic insomnia.¹⁹ Though the exact mechanism is not known, the various factors responsible may include the release of inflammatory mediators like interleukin 6 (IL-6), nuclear factor kappa-B cell (NF- κ B) in chronic insomnia resulting in allergic airway inflammation.^{20,21} It is also noted that chronic insomnia may reduce interferon- γ production that reduces the airway epithelial inflammation and thereby increasing the risk of reactive airway disease in patients with insomnia. Optimal management of chronic insomnia may prevent the release of such inflammatory mediators reducing the risk of airway inflammation.

Insomnia and Thyroid Disorders

The risk of thyroid disorders with chronic insomnia is not very well known. However, studies have shown that the dysregulation in the hypothalamic-pituitary axis in chronic insomnia increases the levels of corticotrophin-releasing hormone (CRH), thyrotropin-releasing hormone (TRH) and cortisol,

resulting in fluctuation in thyroid hormone levels. The abnormal levels of TRH and thyroid stimulating hormone (TSH) are also noted higher in patients with insomnia with comorbid depression.^{[22](#)}

Clinical Assessment

A detailed sleep history is a key to the evaluation of insomnia. Clinicians should be able to recognize the sleep disturbances and rule out other sleep-related disorders like restless leg syndrome, sleep apnea, periodic limb movements, and nocturnal leg cramps that may be contributing to the sleep fragmentation. Complete laboratory workup should be helpful to evaluate any underlying medical conditions contributing to insomnia. Furthermore, questionnaires, sleep logs, and actigraphy can be helpful tools for the assessment of insomnia.

A self-reported questionnaire can be helpful to evaluate the quality of sleep in chronic insomnia. Epworth Sleepiness Scale (total score 0–24; score >15 considered for severe daytime sleepiness) and Pittsburgh Sleep Quality Index (score > five is considered poor sleep score) are the two most widely used assessment tools in doctor's office visits.^{[23](#)}

Sleep diaries are another cost-effective way to evaluate the sleep-wake disturbances in the patients. It is helpful to determine the total sleep time (TST), wakefulness after sleep onset (WASO), sleep efficiency and circadian rhythm disturbances. They also include information about caffeine consumption, medications daytime napping and bedtime activities which are helpful to assess the sleep hygiene in these patients.^{[24](#)}

Wrist actigraphy is a noninvasive tool that records the gross motor activity during sleep and wakefulness. It is useful to estimate the sleep parameters like sleep duration, wakefulness after sleep onset (WASO), sleep latency. Maintaining sleep diaries along with actigraphy can provide complementary information.^{[25](#)} Further, actigraphy is not helpful to assess periodic limb movements or abnormal breathing patterns for which, polysomnography should be chosen. However, polysomnography is not routinely recommended for the initial assessment of insomnia.

Non-Pharmacological Management

1. Sleep Hygiene

Sleep hygiene includes educating the patients about lifestyle modifications like limiting the daytime naps, avoiding late night dinner, restricting the use of electronic gadgets/smartphones during bedtime or evening intake of alcohol, caffeine, or smoking. Certain practice scales like sleep hygiene

index and the sleep hygiene awareness scales are useful to assess the sleep hygiene. However, sleep hygiene alone is ineffective in managing patients with chronic insomnia and should be used with other aspects of cognitive behavior therapy.^{[26](#)}

2. Sleep Restriction Therapy

This therapy aims to reduce sleep time by limiting the number of sleeping hours. Reduced sleep time can improve the homeostatic sleep drive and result in a more consolidated sleep. The major limitation of this therapy is an increased chance of daytime sleepiness due to sleep loss.

3. Stimulus Control Therapy

Stimulus Control involves restriction of maladaptive behaviors like eating or reading in bed, late night use of digital devices in bed and promoting the use of bed for sleeping and only when feeling drowsy.

4. Relaxation Therapy

Regular practice of breathing exercises, meditation or yoga can help to improve the sleeping pattern and reduce underlying anxiety and stress. Studies have shown that management of stress with relaxation and mindfulness training helps to improve focused attention and reduce pre-sleep arousal and worry in insomnia patients.^{[27](#)}

5. Cognitive Behavioral Therapy for Insomnia

Cognitive behavioral therapy for Insomnia (CBTi) is the mainstay of management of insomnia. Effective CBTi can show significant improvement in sleep onset latency (SOL), wakefulness after sleep onset (WASO) and total sleep time (TST). Studies have shown CBTi is superior to pharmacotherapy in the management of chronic insomnia.^{[28](#)} It is typically delivered in six sessions over six- to eight-week period by either health care nurse, sleep therapist, physician assistant, or even a social worker. The sessions include sleep education, relaxation techniques, sleep restriction therapy, stimulus control therapy, cognitive. and behavioral therapy. It can also be provided through the telehealth (video conferencing) or internet-based versions that are beneficial for those who are hesitant to visit a therapist in person. “SHUTi” is an online internet-based CBTi program proven for insomnia. “Sleep Ninja” is a smartphone app, that delivers CBTi over the phone.^{[29](#)} However, the major limitation of these web-based versions is that a lot of self-encouragement is required to follow through the entire length of the programs regularly. Another

limitation of the CBTi program is a shortage of efficient therapists to deliver the therapy effectively along with a higher out of pocket costs, which further restrict the patients from the benefits of the program.³⁰

Pharmacological Management

Drugs Acting on GABA-A Receptors

The benzodiazepines (BZD) and benzodiazepine receptor agonists (BzRA or non-BZD) both act on the gamma-aminobutyric acid (GABA) receptor sites thereby exerting sedative, anxiolytic, muscle relaxant, and hypnotic effects. One significant difference between the two groups is the affinity for different subtypes of GABA alpha subunit. While all the BZD have similar affinity to various subtypes of alpha subunits, BzRA have a varying affinity to different subtypes of alpha subunits. For example, zolpidem, zopiclone, and zaleplon have higher affinity to alpha-1 subunit and lower affinity to alpha-2 and alpha-3 subunit; whereas, eszopiclone has higher affinity to alpha-2 and alpha-3 subunit of GABA receptor.³¹ The adverse effects associated with BZD like rapid development of tolerance, the risk of abuse or dependence, the occurrence of rebound insomnia after drug discontinuation, and cognitive impairment further limit the use of BZD over BzRA

BzRA are approved by the Food and drug administration (FDA) for the management of insomnia. They are rapidly absorbed, relatively short-acting (as compared to benzodiazepines) and have better side effect profiles. They are effective in treating sleep onset insomnia, sleep maintenance insomnia or both.

Zolpidem binds selectively to the alpha-one subtype of GABA-A receptor. It has a short half-life of 2.5 hours and is available in immediate-release (IR) formulation of 5-mg and 10-mg doses, which are effective for the treatment of short-term insomnia. The controlled-release (CR) form is available in 6.25-mg and 12.5-mg dosage for sleep onset and sleep maintenance insomnia. A sublingual form (doses in male 3.5 mg and female 1.75 mg) is available for the treatment of middle of night awakenings and difficulty in returning to sleep and should be used if there is a minimum of 4 or more hours of intended sleep time. The adverse effects associated with zolpidem are headache, falls, somnolence, and antegrade amnesia. ([Table 1](#)).

Table 1.

The table summarizes the drug therapy approved for chronic insomnia along with their brand names and half-life elimination. (GABA: gamma-aminobutyric acid; BzRA: Benzodiazepine receptor agonist)

Drugs acting on various receptors	Elimination half-life(hour)
GABA receptors	
Benzodiazepines	
Triazolam (Halcion [®])	1.5–5.5
Temazepam (Restoril [®])	3.5–18.4
Estazolam (ProSom [®])	10–24
Flurazepam (Dalmane [®])	48–120
BzRA	
Zolpidem -oral (Ambien [®])	2.5
Zolpidem- oral spray (Zolpimist [®])	2.8
Zolpidem-extended release (Ambien CR [®])	2.8(1.6–4.5)
Zolpidem -Sublingual(Intermezzo [®])	2.5
Eszopiclone (Lunesta [®])	6–9
Zaleplon (Sonata [®])	1
Histamine-1 receptor antagonist	
Doxepin (Silenor [®])	15.3
Melatonin receptor agonist	
Ramelteon (Rozerem [®])	1–2.6
Orexin receptor antagonist	
Suvorexant (Belsomra [®])	12

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Zaleplon has the shortest duration of action with the half-life of one hour and is available at the doses of 5 mg, 10 mg, 20 mg for the treatment of insomnia. The adverse effects associated with it are a headache, drowsiness, nausea, and worsening of depressive symptoms in patients with the comorbid depressive disorder.

Eszopiclone helps to improve sleep efficiency, daytime functioning along with a reduction in sleep onset latency and wakefulness after sleep onset. It is used for management of sleep onset insomnia (2 mg) and sleep maintenance (3 mg) insomnia. It acts on the alpha-2, and alpha-3 receptors subtype of the GABA-A receptors, thereby exerting anxiolytic and antidepressant effect respectively, and hence, is effective in the management of insomnia with comorbid depression or generalized anxiety disorder. Common adverse effects associated with eszopiclone are unpleasant metallic taste, headache, dizziness, and somnolence.³²

Drugs Acting on Melatonin Receptors

Melatonin is a natural hormone produced by the pineal gland. The circadian system in the hypothalamus and the suprachiasmatic nucleus (SCN) regulates the levels of this hormone throughout the day and night. Melatonin is available over the counter and is approved by FDA for treatment of insomnia, especially in older adults. A dose range of 2 to 8 mg is effective in treating circadian rhythm sleep-wake disorders. However, food can delay the absorption of melatonin, and a gap should be maintained between the last meal of the day and the intake of melatonin.

Ramelteon, melatonin receptor agonist decreases the sleep latency by acting on the melatonin MT1 and MT2 receptors in the SCN with higher affinity than melatonin itself.³³ The FDA recommends a dosage of 8 mg for the management of sleep onset insomnia. It exerts minimal adverse effects including somnolence, fatigue, and dizziness.

Tasimelteon is another melatonin receptor agonist effective in improving sleep initiation and maintenance particularly in blind patients with Non-24-hour sleep-wake circadian rhythm disorders.³⁴

Drugs Acting as Orexin Receptor Antagonist

Suvorexant is a dual orexin receptor antagonist (OX1 and OX2 receptor) which counteracts the orexin/hypocretin system that plays an important role in wakefulness. It is effective in doses of 5 mg, 10 mg, 15 mg, and 20 mg for the management of sleep onset and sleep maintenance insomnia. A dose of 15 mg and 20 mg has shown improvement in total sleep time and a reduction in sleep

onset latency. However, the FDA does not recommend a higher dose of 30 mg or 40 mg of suvorexant because of safety concerns, with an increased risk of next day driving difficulty, increased daytime somnolence and narcolepsy-like symptoms (hypnogogic-hypnopompic hallucinations, cataplexy, and vivid dreams). Also, suvorexant is contraindicated in patients with narcolepsy because of possible underlying mechanisms of orexin antagonism.³⁵

Drugs Acting as Histamine-1 Receptor Antagonist

Doxepin is a tricyclic antidepressant, but at a low dose of 3 mg and 6 mg, it is effective in the management of sleep maintenance insomnia. It causes improvement in total sleep time, wakefulness after sleep onset and sleep efficiency. At low doses (3 mg and 6 mg), Doxepin acts as pure H-1 receptor antagonist being 800 times more potent than diphenhydramine for H-1 receptors, and at high doses of 25 mg to 300 mg daily (antidepressant dosage), it exerts antihistaminic, antiserotonergic, anticholinergic and antiadrenergic activity. The adverse effects associated with doxepin at low doses are headache and somnolence.³⁶

Off-Label Drugs

1. Antidepressants: trazodone, mirtazapine, and amitriptyline are most commonly used antidepressants for the management of insomnia at low doses mainly because of their antihistaminic effect. Studies have shown a 50 mg once a day dose of trazodone has proved to be effective in improving sleep latency, wakefulness after sleep onset and duration of sleep.³⁷
2. Atypical antipsychotics: Olanzapine and quetiapine can be useful in the treatment of insomnia with comorbid psychotic conditions. They exert a sedative effect at low doses mainly by their antihistaminic, anti-adrenergic and antidopaminergic properties.³⁸
3. Anticonvulsants: Gabapentin has shown to improve the sleep efficiency and decrease the wakefulness after sleep onset. It can be effective in managing insomnia in patients with alcohol dependence. Pregabalin increases the total sleep time, stage N3, sleep efficiency and decreases sleep onset latency and REM sleep. It is helpful in improving sleep in patients with generalized anxiety disorder and fibromyalgia.³⁹

The American Academy of Sleep Medicine (AASM) has proposed a PICO (Patient, population, problem, Intervention, Comparison, and Outcomes) template, based on the patient-oriented tools to determine the outcomes in response to the treatment given for insomnia. The four most critical outcomes

useful for clinical decision-making are sleep latency (SL), wake after sleep onset (WASO), total sleep time (TST), and the quality of sleep (QoS) ([Figure 3](#)).⁴⁰

Figure 3.

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Effect of various pharmacological agents on Total Sleep time (TST), Wakefulness after sleep onset (WASO), Sleep latency (SL), Quality of Sleep (QOS).

Conclusion

Chronic insomnia is a rising public health concern affecting the quality of life of up to one-third of the population of the United States. It can be associated with psychopathology along with a variety of systemic disorders. Genetic and environmental factors play a role in the pathogenesis of this problem. Cognitive behavioral therapy is still the first line treatment for insomnia though the scarcity of therapists and the high costs make it less practical in many cases. Pharmacotherapy can be effective in many patients and should always be used in conjunction with sleep hygiene.

Footnotes

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Disclosure

None reported.

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